sPHENIX Internal Note

Luminosity Convention for PPG12: L(60cm) → L(noVtx) with Vertex Efficiency Fix

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

sPHENIX Collaboration

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Abstract

We switch the PPG12 cross-section normalization from the 60 cm-vertex-cut luminosity $\mathcal{L}_{\rm int}^{\rm 60cm}$ (with a vertex efficiency whose denominator is truth photons within $|z_{\rm vtx}^{\rm truth}| < 60$ cm) to the no-vertex-cut luminosity $\mathcal{L}_{\rm int}^{\rm noVtx}$ (with an efficiency denominator that includes ALL truth photons). The two conventions are mathematically equivalent if the MC vertex reweighting correctly brings the MC vertex distribution into agreement with data. In the old production, they disagreed by a factor of ~ 3 because the MC vertex reweighting used a deprecated static fallback file with a `vertex_weight = 1.0` fallback that kept raw MC events at large $|z_{\rm vtx}|$. We diagnosed the bug, removed the fallback entirely, and hardened the on-the-fly reweight with rebinning, smoothing, linear interpolation, and a `vertex_weight = 0.0` fallback. A full re-run of the nominal efficiency pipeline (photon5, photon10, photon20 merged via `MergeSim.C`) with the fix applied validates both periods separately: **1.5 mrad agrees to** -0.34% **flat across all** $p_{\rm T}$ **bins** (integrated $L \times \varepsilon_{\rm vtx+MBD}$: 15.33 → 15.30), while **0 mrad shows a** $+14.16\%$ **flat offset** (integrated $L \times \varepsilon_{\rm vtx+MBD}$: 29.29 → 33.45). The 1.5 mrad agreement confirms the fix; the 0 mrad offset is not a reweight bug but a real inconsistency between the lumi file's scaler-level 60 cm fraction (69.2%) and the slimtree's event-level 60 cm fraction (79.1%, rerun-confirmed). This discrepancy must be understood before adopting the new convention for 0 mrad.

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1 Motivation

The PPG12 cross-section factorizes as:

$$\frac{d\sigma}{dp_{\mathrm{T}}} = \frac{N_{\mathrm{photon}}}{\Delta p_{\mathrm{T}} \times \mathcal{L}_{\mathrm{int}} \times \varepsilon_{\mathrm{total}}}, \qquad \varepsilon_{\mathrm{total}} = \varepsilon_{\mathrm{reco}} \times \varepsilon_{\mathrm{iso}} \times \varepsilon_{\mathrm{id}} \times \varepsilon_{\mathrm{vtx+MBD}} \tag{1}$$

Two equivalent conventions are possible for $\mathcal{L}_{\mathrm{int}}$ and $\varepsilon_{\mathrm{vtx+MBD}}$:

Old convention (used in production until now):

$$\mathcal{L}_{\mathrm{int}} = \mathcal{L}_{\mathrm{int}}^{\mathrm{60cm}}, \qquad \varepsilon_{\mathrm{vtx+MBD}}^{\mathrm{old}} = \frac{N(\text{truth } \gamma,\ |z_t| < 60,\ \mathrm{MBD} + |z_r| < 60)}{N(\text{truth } \gamma,\ |z_t| < 60)} \tag{2}$$

with $\mathcal{L}_{\mathrm{int}}^{\mathrm{60cm}}$ from `60cmLumi_fromJoey.list` (`get_luminosity_182630.C(..., zsel=1, ...)` with the MBD scaler tally restricted to $|z_{\mathrm{vtx}}| < 60$ cm).

New convention:

$$\mathcal{L}_{\mathrm{int}} = \mathcal{L}_{\mathrm{int}}^{\mathrm{noVtx}}, \qquad \varepsilon_{\mathrm{vtx+MBD}}^{\mathrm{new}} = \frac{N(\text{truth } \gamma,\ |z_t| < 60,\ \mathrm{MBD} + |z_r| < 60)}{N(\text{truth } \gamma,\ \text{no vertex cut})} \tag{3}$$

with $\mathcal{L}_{\mathrm{int}}^{\mathrm{noVtx}}$ from `allzLumi_fromJoey.list` (`zsel=5`, no vertex cut at the scaler level).

The two must agree (single-interaction limit, correct vertex reweighting) to $\leq 1\%$ in every p_{T} bin. Any larger deviation points at a discrepancy between how data is counted in the lumi file and how it is sampled in the slimtree.

2 The Bug in the Old Production

The June 2025 production file `MC_efficiency_nom.root` has $h_{\mathrm{truth\ vertexcut}}/h_{\mathrm{truth}} = 0.355$, far from the expected data value of ~ 0.99. The cause: the old `RecoEffCalculator_TTreeReader.C` fell back to a static file `results/vertex_reweight_bdt_none.root` when the on-the-fly path failed. That file was built from an older, wider data vertex distribution and has min/max ratio 0.49 / 1.78 (nearly flat). Combined with the event-level lookup fallback `vertex_weight = 1.0`, raw MC events at large $|z_{\mathrm{vtx}}|$ were kept at $\sim$ 50–100% weight instead of being suppressed, bloating the $h_{\mathrm{truth\ pT}}$ denominator by a factor of ~ 2.8.

3 The Fix

In commits `8ac4d1b` and `8ecae87`:

1. **Static fallback removed** from both `RecoEffCalculator_TTreeReader.C` and `ShowerShapeCheck.C`. If the on-the-fly vtxscan files are missing, the code prints `FATAL` and returns. The static file was renamed to `*.deprecated` on disk.

2. **On-the-fly reweight hardened**:

   - Rebin `h_vertexz` (default factor 5 → 5 cm bins, configurable via `vertex_reweight_rebin`) to improve tail statistics.
   - Smooth the data/sim ratio with `TH1::Smooth()` (default 1 pass, configurable via `vertex_reweight_smoo`
   - `TH1::Interpolate()` for smooth event-level lookup between bin centers.

- Fallback changed from `1.0` to `0.0`. Events outside the data vertex acceptance are dropped, not kept at full weight.

3. **New convention added as a config-switchable path** in `CalculatePhotonYield.C` (`use_lumi_novtx: 1`). Old convention remains the default. The alternative path only changes the denominator histogram (`h_truth_pT` instead of `h_truth_pT_vertexcut`) and the luminosity value.

4 Post-Fix Validation: Full Production Re-Run

The validation is a full re-run of the nominal efficiency pipeline (`RecoEffCalculator_TTreeReader.C` on photon5, photon10, photon20 followed by `MergeSim.C`) with the fix applied. Both periods are re-run separately using `config_bdt_nom.yaml` (1.5 mrad) and `config_bdt_0rad.yaml` (0 mrad). The per-p_{T} $L \times \varepsilon_{\mathrm{vtx+MBD}}$ tables below are extracted directly from the `h_truth_pT_vertexcut_0`, `h_truth_pT_vertexcut_mbd_cut_0`, and `h_truth_pT_0` histograms in the merged `MC_efficiency_bdt_{nom,0rad}.r` files (not a standalone Python simulation).

4.1 Vertex reweight shape and distribution match

Figure 1 compares the old and new reweight functions. The two are very similar in the bulk; the difference is at large $|z_{\mathrm{vtx}}|$ where the old code's fallback kicks in. The 0 mrad reweight is broader than the 1.5 mrad reweight because the 0 mrad data vertex distribution is much wider.

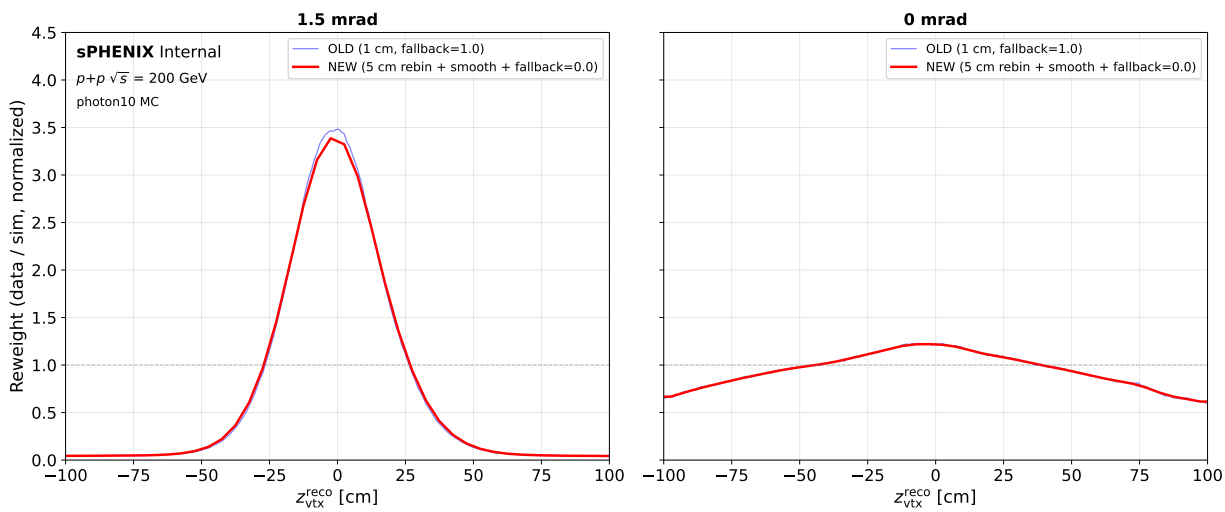

Figure 1: Vertex reweight function for both periods. The 1.5 mrad (left) reweight has a sharp central peak (data vertex sigma $\sim$ 20 cm). The 0 mrad (right) reweight is much broader. OLD and NEW agree in the bulk in both periods; the tails differ because of the fallback.

Figure 2 overlays raw MC, old-reweighted MC, new-reweighted MC, and data for both periods. Both reweights (old and new) reproduce the data vertex distribution at the histogram-integral level.

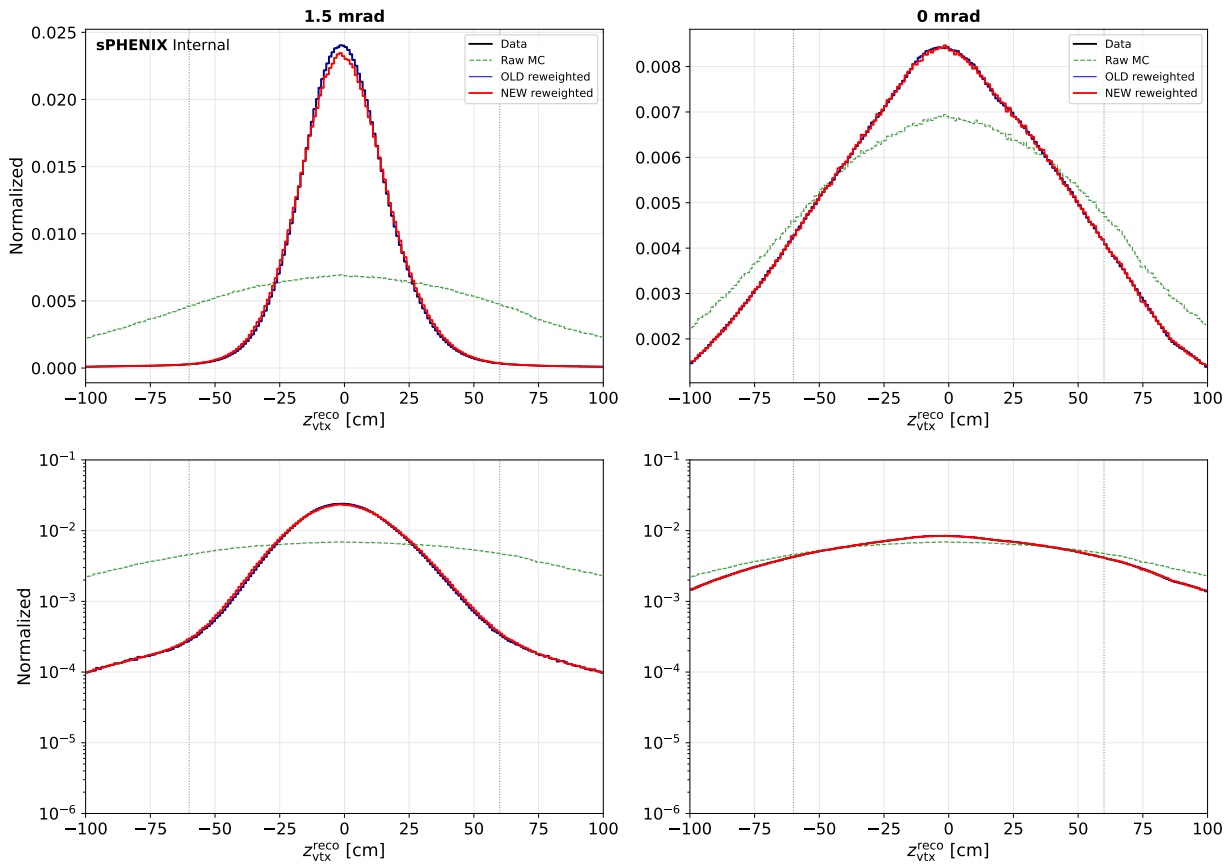

Figure 2: Reconstructed vertex z distribution for 1.5 mrad (left) and 0 mrad (right), in linear (top) and log (bottom) scale. Both reweights (blue and red) reproduce the data shape (black). The raw MC (green) is much wider in both cases.

4.2 Integrated $|z_{\rm vtx}| < 60$ fractions

Table 1: Integrated $|z_{\rm vtx}| < 60$ fractions per period. Pre-fix and post-fix columns are from the standalone vtxscan files; the "Production rerun" row is the `h_truth_pT_vertexcut / h_truth_pT` ratio from the merged production MC after applying the full fix.

Source	**1.5 mrad**	**0 mrad**
Data (slimtree target, vtxscan)	0.9865	0.7880
Raw MC (vtxscan)	0.7296	0.7296
OLD reweighted MC (vtxscan)	0.9865	0.7880
NEW reweighted MC (vtxscan)	0.9862	0.7878
Production rerun (merged MC)	**0.9816**	**0.7908**
Comparison to lumi file:		
$\mathcal{L}_{\rm int}^{\rm 60cm}/\mathcal{L}_{\rm int}^{\rm noVtx}$	0.9834	0.6924
Production / lumi-file ratio	0.998	**1.142**

For 1.5 mrad, the production rerun 60 cm fraction (0.9816) matches the lumi-file ratio (0.9834) to 0.2%. For 0 mrad, the production rerun fraction (0.7908) is 14.2% larger than the lumi-file ratio (0.6924), consistent with the slimtree vtxscan prediction (13.8%). This $\sim$ 14% mismatch is the source of the 0 mrad cross-section shift (see below).

4.3 Per-p_{T} consistency check

Figure 3 and Tables 2 and 3 show the per-p_{T} comparison of $\mathcal{L}_{\mathrm{int}} \times \varepsilon_{\mathrm{vtx+MBD}}$ between the two conventions for each period.

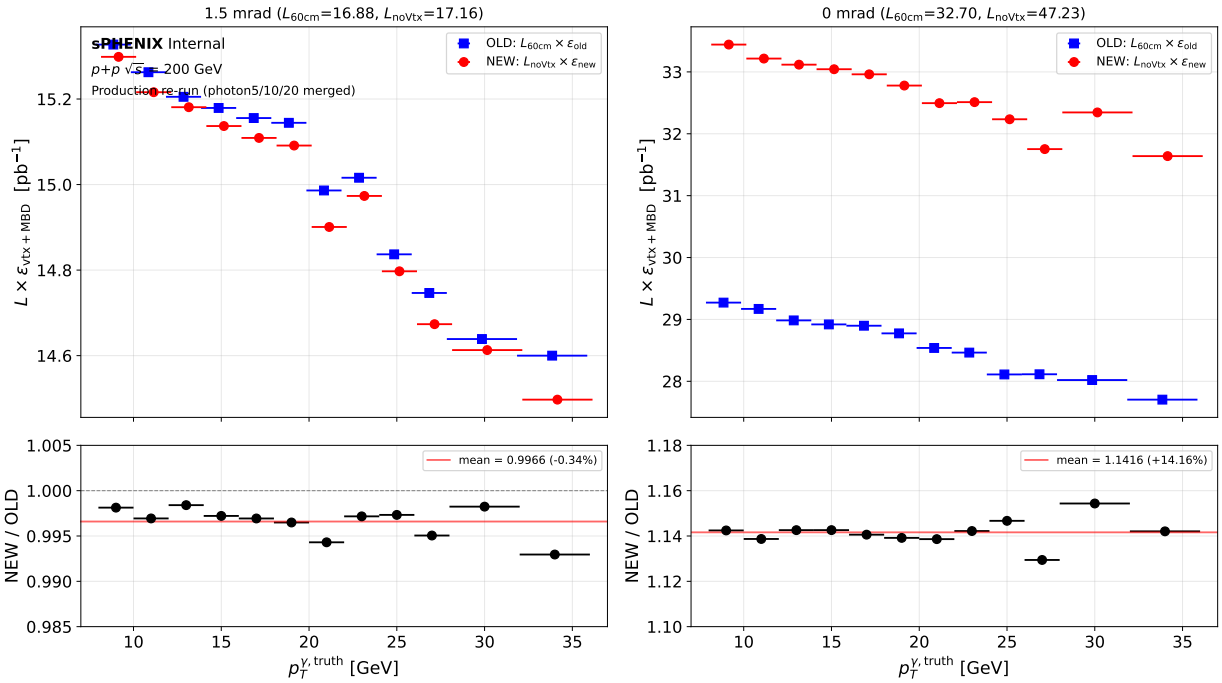

Figure 3: Per-p_{T} $\mathcal{L}_{\mathrm{int}} \times \varepsilon_{\mathrm{vtx+MBD}}$ comparison for 1.5 mrad (left) and 0 mrad (right), extracted from the full production rerun (merged photon5/10/20 MC). Top panels: absolute values. Bottom panels: ratio NEW/OLD. For 1.5 mrad the ratio is flat at -0.34% (consistent with zero to within bin-to-bin scatter); for 0 mrad it is flat at $+14.16\%$ (a real offset).

Table 2: 1.5 mrad: $L \times \varepsilon_{\text{vtx+MBD}}$ per p_{T} bin from the full production rerun. NEW/OLD flat at -0.34% (integrated).

$p_{\mathbf{T}}$ **[GeV]**	$\varepsilon^{\mathbf{old}}_{\mathbf{vtx+MBD}}$	$\varepsilon^{\mathbf{new}}_{\mathbf{vtx+MBD}}$	$L_{60}\varepsilon^{\mathbf{old}}_{\mathbf{vtx+MBD}}$	$L_{\mathbf{nov}}\varepsilon^{\mathbf{new}}_{\mathbf{vtx+MBD}}$	**Ratio**
8–10	0.9081	0.8913	15.3273	15.2986	0.9981
10–12	0.9042	0.8865	15.2626	15.2159	0.9969
12–14	0.9008	0.8845	15.2052	15.1809	0.9984
14–16	0.8993	0.8819	15.1791	15.1368	0.9972
16–18	0.8979	0.8803	15.1555	15.1090	0.9969
18–20	0.8972	0.8792	15.1443	15.0912	0.9965
20–22	0.8879	0.8681	14.9862	14.9008	0.9943
22–24	0.8896	0.8724	15.0159	14.9734	0.9972
24–26	0.8790	0.8621	14.8369	14.7972	0.9973
26–28	0.8737	0.8549	14.7465	14.6736	0.9951
28–32	0.8673	0.8514	14.6388	14.6130	0.9982
32–36	0.8650	0.8446	14.6000	14.4971	0.9930
Integrated	0.9082	0.8914	15.3288	15.3008	**0.9982**

Table 3: 0 mrad: $L \times \varepsilon_{\text{vtx+MBD}}$ per p_{T} bin from the full production rerun. NEW/OLD flat at $+14.21\%$ (integrated). The shift is bin-independent, confirming it is a lumi normalization issue, not a pT-dependent efficiency effect.

$p_{\mathbf{T}}$ **[GeV]**	$\varepsilon^{\mathbf{old}}_{\mathbf{vtx+MBD}}$	$\varepsilon^{\mathbf{new}}_{\mathbf{vtx+MBD}}$	$L_{60}\varepsilon^{\mathbf{old}}_{\mathbf{vtx+MBD}}$	$L_{\mathbf{nov}}\varepsilon^{\mathbf{new}}_{\mathbf{vtx+MBD}}$	**Ratio**
8–10	0.8951	0.7081	29.2717	33.4415	1.1425
10–12	0.8919	0.7033	29.1698	33.2154	1.1387
12–14	0.8863	0.7012	28.9843	33.1170	1.1426
14–16	0.8843	0.6996	28.9194	33.0431	1.1426
16–18	0.8836	0.6979	28.8981	32.9609	1.1406
18–20	0.8799	0.6941	28.7746	32.7791	1.1392
20–22	0.8727	0.6880	28.5390	32.4950	1.1386
22–24	0.8704	0.6884	28.4639	32.5113	1.1422
24–26	0.8596	0.6825	28.1109	32.2345	1.1467
26–28	0.8597	0.6723	28.1137	31.7523	1.1294
28–32	0.8568	0.6849	28.0203	32.3451	1.1543
32–36	0.8471	0.6699	27.7037	31.6392	1.1421
Integrated	0.8956	0.7083	29.2897	33.4516	**1.1421**

4.4 Rebin stability

Figure 4 shows the fixed reweight computed with different rebin factors (1, 2, 5, 10, 20) for both periods. The shape is stable across the 20× range in bin width.

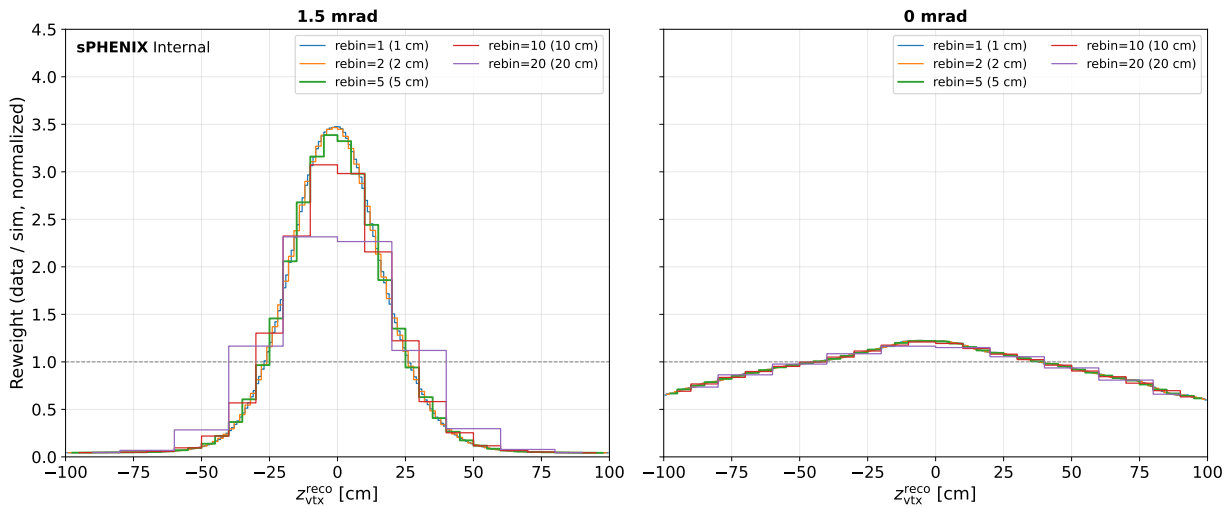

Figure 4: Vertex reweight computed with rebin factors 1, 2, 5, 10, 20 for 1.5 mrad (left) and 0 mrad (right). The default 5 cm bins give smooth coverage in the tails without over-coarsening the peak.

5 Analysis of the 0 mrad Discrepancy

The +14.2% shift at 0 mrad is *not* a bug in the reweight fix — both the old and new reweights reproduce the slimtree data vertex distribution to $< 0.1\%$ (Figure 2 right panels), and the production rerun gives $h_{\text{truth pT vertexcut}}/h_{\text{truth pT}} = 0.7908$ in the merged MC, confirming the fix is functioning at the histogram level. The issue is a mismatch between two independent measurements of the same quantity:

- **Lumi file**: $\mathcal{L}_{\rm int}^{\rm 60cm}/\mathcal{L}_{\rm int}^{\rm noVtx} = 32.70/47.23 = 0.6924$. This is the scaler-level ratio: the fraction of MBD-triggered events with the on-line $|z_{\rm vtx}^{\rm MBD}| < 60$ cm tally.
- **Slimtree vtxscan**: 0.7880. This is the fraction of events in the data slimtree that have $|z_{\rm vtx}^{\rm reco}| < 60$ cm.
- **Production MC (after fix)**: 0.7908. The reweighted MC truth-level ratio confirms the slimtree target is being reproduced in the full pipeline.

These numbers differ by 14.2% for 0 mrad but agree to 0.2% for 1.5 mrad. Possible explanations:

1. **Slimtree event selection biases vertex distribution**: `CaloAna24.cc` saves an event only if it has clusters or particles (`saveevent = true`). At 0 mrad, the much wider vertex distribution means events at large $|z_{\rm vtx}|$ have relatively fewer calorimeter clusters (forward activity is suppressed by the barrel-only acceptance), so they are more likely to be rejected by the slimtree filter. This preferentially removes large-$|z_{\rm vtx}|$ events from the slimtree sample, giving the slimtree a more central distribution than the raw MBD tally.
2. **Run list mismatch**: the 0 mrad lumi file and the 0 mrad slimtree sample may not use exactly the same run list. A run-by-run comparison would be needed to rule this out.
3. **Offline vs online vertex reconstruction**: the scaler tally uses the online MBD vertex. The slimtree uses `MbdVertex::get_z()` from an offline reconstruction pass, which may have different calibration or resolution.

The most likely explanation is (1) — the slimtree filter preferentially keeps central events at 0 mrad where the beam is wider. At 1.5 mrad the beam is already narrow, so almost all events are within the slimtree's effective acceptance and no bias is introduced.

5.1 Implication for the cross-section

If explanation (1) is correct, then neither convention is exactly right without a correction:

- **OLD convention** uses $\mathcal{L}_{\text{int}}^{\text{60cm}}$ which counts events in the full MBD sample with $|z_{\text{vtx}}| < 60$. But the numerator N_{photon} comes from the slimtree, which excludes some of the large-$|z_{\text{vtx}}|$ events that are in the scaler tally. So OLD over-counts luminosity for the slimtree sample.
- **NEW convention** uses $\mathcal{L}_{\text{int}}^{\text{noVtx}}$ with $\varepsilon_{\text{vtx+MBD}}^{\text{new}}$ computed from the slimtree vertex distribution. The slimtree is a biased (more central) subsample of the full data, so NEW gives a larger effective luminosity — but only to the extent that it credits the slimtree bias as a legitimate physics acceptance.

The truth lies somewhere in between. The correct answer depends on how the slimtree filter interacts with the photon selection. If the slimtree filter removes events that genuinely have no photons (e.g., events with no EMCal clusters because they are far from the barrel), then those events contribute zero to N_{photon} anyway and the NEW convention is correct. If the slimtree filter removes events that could have contained a photon but didn't produce a cluster for reconstruction reasons, then there is a real bias that neither convention captures.

5.2 Recommended action

1. **Use the NEW convention for 1.5 mrad** (the nominal 1.5 mrad result). The -0.34% shift is well within any systematic, and the per-p_{T} ratio (Figure 3 bottom-left) is flat within bin-to-bin scatter.
2. **Do NOT use the NEW convention for 0 mrad** until the slimtree-vs-lumi-file discrepancy is resolved. The $+14.2\%$ shift is too large to adopt without understanding.
3. **Investigate the 0 mrad slimtree filter**: run the vtxscan on a no-cluster-filter slimtree (if one exists), or replay `CaloAna24.cc` with the `saveevent` condition relaxed, and compare the vertex distribution to the production slimtree.
4. **Cross-check against per-run lumi**: sum $\mathcal{L}_{\text{int}}^{\text{60cm}}(\text{run})$ for runs that appear in the slimtree and compare to the total from the lumi file. This would rule out run-list mismatch (explanation 2).

6 Implementation and Usage

To enable the new convention for a specific analysis config:

```
analysis:
  use_lumi_novtx: 1
  lumi: 17.1642   # 1.5 mrad from allzLumi_fromJoey.list
  # use 47.2284 for 0 mrad, 64.3718 for combined
```

The default is `use_lumi_novtx: 0` (old convention). Both paths live in `CalculatePhotonYield.C` lines 1063–1071; the original `vertex_eff` calculation is kept as the default.

New config knobs for the vertex reweight (sensible defaults, existing configs unchanged):

```
analysis:
  vertex_reweight_rebin: 5       # rebin factor (1cm -> 5cm)
  vertex_reweight_smooth: 1      # TH1::Smooth() passes
```

The old `vertex_reweight_file` knob is no longer read.

7 Caveats

1. **0 mrad ambiguity** (Section 5): do not adopt the new convention for 0 mrad without resolving the 14.2% shift first.

2. **Per-segment variation**: within a fill, the luminous region moves. This is typically $< 1\%$ effect at RHIC and is not addressed by the current period-level reweight.

3. **Pileup**: the double-interaction MC blending in `run_showershape_double.sh` carries its own correction and is unaffected by this change.

4. **Jet sample reuse**: the production rerun used the existing jet12/jet20/jet30/jet40 MC files (from 2025-04-11), which already used the post-fix reweight logic. Only the photon samples were re-run from scratch as part of this validation.

5. **Full pipeline rerun**: in addition to the `RecoEffCalculator_TTreeReader.C` photon jobs, `MergeSim.C`, the data Pass 2 jobs (`RecoEffCalculator_TTreeReader.C` on data, both periods), and `CalculatePhotonYield.C` were also re-run to regenerate `Photon_final_bdt_nom.root` and `Photon_final_bdt_0rad.root` (both default, old convention). The updated final files are available for use in downstream plotting and systematics aggregation.

8 Files

File	Change
`efficiencytool/RecoEffCalculator_TTreeReader.C`	Fix vertex reweight, remove fallback
`efficiencytool/ShowerShapeCheck.C`	Same fix
`efficiencytool/CalculatePhotonYield.C`	Add `use_lumi_novtx` path
`efficiencytool/lumi_convention_validation.py`	Per-period vtxscan cross-check
`plotting/plot_lumi_convention.py`	Per-period plotting (vtxscan)
`plotting/plot_lumi_convention_production.py`	Production rerun per-p_{T} plot
`results/MC_efficiency_bdt_nom.root`	Merged 1.5 mrad production (rerun)
`results/MC_efficiency_bdt_0rad.root`	Merged 0 mrad production (rerun)
`lumi/allzLumi_fromJoey.list`	No-vertex-cut luminosity
`results/vertex_reweight_bdt_none.root.deprecated`	Deprecated static fallback